

KIRSH KUMAR

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SUMMARY

: **Final-year** Computer Science student at FAST-NUCES Karachi with a strong interest in Artificial Intelligence, software development, and digital media. Experienced in building AI and CS-based projects and actively contributing to university societies in roles such as Media Manager, Web Developer, and Marketing Lead.

EDUCATION

FAST NATIONAL UNIVERSITY OF COMPUTER SCIENCE

August 2023 - June 2027

Bachelor's in Computer Science | GPA: 3.60

Relevant Coursework: Programming Fundamentals | Object-Oriented Programming | Data Structures | Operating Systems | Databases | Software Design | Artificial Intelligence | Recommender System | Software Engineering | Information Retrieval

Awards: 6x Dean's List

EXPERIENCE

.dot — Software Engineering Intern | June 2026 – August 2026

- Worked on InkMimic, an AI-powered handwriting generation platform that replicates individual handwriting styles using a stroke-level diffusion model.
- Contributed to the AI/ML pipeline, backend APIs, and web/mobile application development.
- Worked with Python, PyTorch, TensorFlow, FastAPI, React, React Native, and Supabase to build and integrate core product functionality.

The Bridge of Hopes Research Lab — AI Intern | July 2026 – September 2026

- Worked on **AI-Enabled Adaptive Learning Systems for Special Needs Children**.
- Designed and developed lightweight AI models for mobile application deployment.
- Conducted literature reviews and designed AI pipelines for model development, evaluation, optimization, and deployment.
- Trained, validated, and improved machine learning and deep learning models.

Nyrix Technologies — Content Marketing Intern | July 2026 - Present

- Created and managed digital content and marketing material to support brand visibility and online engagement.
- Collaborated with team members to develop creative campaigns and promotional content.

PROJECTS

InkMimic — AI Handwriting Generation & Personalization System — Python + FastAPI + React + React Native + TensorFlow/PyTorch + Supabase Stack

- An AI-powered handwriting generation platform that transforms handwriting samples into realistic handwritten text, using a stroke-level diffusion model with style transfer and handwriting analysis. The system supports typed-to-handwriting generation, assignment-length page rendering, handwriting style analysis, high-quality ink rendering, and web/mobile interfaces, powered by a FastAPI backend, React/Vite web app, Expo React Native mobile app, and Supabase for authentication and data management.

SwiftBus - Intercity Bus Management System — React + TypeScript + Supabase Stack

- A full-stack intercity bus booking and management platform with a React and TypeScript UI, Supabase-powered backend, and PostgreSQL database — built for secure authentication, real-time bus tracking, seat booking, digital tickets, payments, refunds, and role-based portals for passengers, drivers, and administrators.

CGAN-based movie recommender system — Python, PyTorch, Streamlit, shell

- Developed a **CGAN-based recommender system** using the MovieLens 100K dataset, modeling user-item interactions through adversarial learning. Implemented generator and discriminator networks for personalized recommendation, human-readable outputs, and cold-start support via interactive user rating.

WanderLust — Travel Web App — MEN - Stack

- WanderLust is a full-stack travel web app built using EJS and CSS for the frontend, Node.js and Express.js for the backend, and MongoDB for data storage. It follows the MVC architecture for clean code separation and integrates Cloudinary for image uploads, Mapbox for dynamic location mapping, and Passport.js for secure authentication and session management. The app supports full CRUD functionality with role-based access control.

SKILLS

Languages: C, C++, Python, HTML, CSS, JavaScript

AI & Machine Learning: TensorFlow, PyTorch, OpenCV, scikit-image, Scikit-learn, NumPy, Pandas, MobileNetV2, Deep Learning, Computer Vision, Generative AI, NLP

Web & Mobile Development: React.js, React Native, Expo, Node.js, Express.js, FastAPI, Bootstrap, Tailwind CSS, REST APIs

Databases & Backend: PostgreSQL, Supabase, MySQL, SQL, Authentication, Database Design

Soft Skills: Communication, Problem-Solving, Multitasking, Teamwork, Graphic Design, Video Editing

Creative Tools: Canva, Photoshop, CapCut, Illustrator (intermediate).

AI & Developer Tools: Claude, Cursor, Antigravity, Git, GitHub, Streamlit, Jupyter Notebook, VS Code